# Automated Anatomy-Grounded Lesion Captioning for Whole-Body PSMA PET/CT Using Large Language Models

Mahdi Shakouri[1,2]

[1]University of British Columbia, Vancouver, BC, Canada
[2]BC Cancer, Vancouver, BC, Canada

## Abstract for Technical Review

Manual lesion-level annotation of whole-body $^{68}$Ga/$^{18}$F-PSMA PET/CT studies is labor-intensive and represents a bottleneck for curating datasets to train and evaluate vision-language models in nuclear medicine. We present an end-to-end, five-stage pipeline that converts free-text PSMA PET/CT reports and voxel-level lesion segmentations into automatically generated, anatomically grounded, lesion-level captions, together with a multi-metric evaluation framework. Structured SUVmax mentions were first extracted from 286 clinical reports (598 mentions across four organ-system sections). Lesion masks were labeled via 26-connectivity connected-component analysis, and per-lesion SUVmax, three-dimensional centroid, metabolic tumor volume (MTV), and bounding-box dimensions were computed from co-registered PET volumes. Because absolute SUVmax values in the anonymized PET data diverged systematically (patient-specific scale drift, median $\sim$4.3$\times$) from the values in the original report text, report-to-lesion correspondence was established via rank-order matching of SUVmax values within each patient rather than absolute-value thresholding, exploiting well-preserved rank concordance. Whole-body CT volumes were segmented into 117 anatomical structures using TotalSegmentator, and for each lesion the three nearest anatomical structures were identified via nearest-neighbor search in world coordinates. A structured prompt encoding lesion centroid, SUVmax, MTV, dimensions, and the three nearest anatomic structures (with distances and centroids) was submitted to a large language model role-framed as a nuclear medicine physician, producing free-text lesion descriptions. Generated captions were evaluated against ground-truth report text using BARTScore, BERTScore, RaTEScore, and GREEN. In an interim analysis of 384 lesions (from 232 of 286 patients with completed CT segmentation), the pipeline achieved a 79.9% report-to-lesion correspondence rate and produced captions with a mean BERTScore F1 of 0.872 and RaTEScore of 0.481 against ground-truth report text. This work demonstrates a scalable, reproducible approach to lesion-level caption generation that avoids direct image-token LLM input by grounding generation in structured, automatically derived anatomic and quantitative features.

**Keywords:** PSMA PET/CT, prostate cancer, nuclear medicine, lesion captioning, large language models, radiology report generation, automated anatomic segmentation, natural language generation evaluation, multimodal AI, structured reporting

## Summary for Program

We built an automated pipeline that reads PSMA PET/CT lesion data and anatomical context to generate free-text radiology-style descriptions of individual tumor lesions using a large language model, then scored the generated text against the original clinical reports using multiple automated language-quality metrics.